RUIJUN DANG

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EDUCATION

Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China 2015 – 2020
Ph. D., Atmospheric Physics and Atmospheric Environment

Nanjing University, Nanjing, Jiangsu, China 2011 – 2015
B.S., Atmospheric Sciences
Minor in Finance

PROFESSIONAL EXPERIENCE

Assistant Professor, Peking University May 2026 – Present
School of Earth and Space Sciences

Research Associate, Harvard University 2024 – 2026
School of Engineering and Applied Sciences

Postdoctoral Research Fellow, Harvard University 2021 – 2024
Advisor: Daniel J. Jacob

Research Assistant, Institute of Atmospheric Physics 2020 – 2021
Advisor: Hong Liao

Graduate Student Researcher, Institute of Atmospheric Physics 2015 – 2020
Advisor: Hong Liao

Undergraduate Researcher, Nanjing University 2014 – 2015
Advisor: Jane Liu

Undergraduate Researcher, Nanjing University 2013 – 2014
Advisor: Tijian Wang

PUBLICATIONS

h-index = 17, total citations = 1253 (as of September 2026 on Google Scholar)

In review:

1. He, X., Chen, Z., Van der A, R., Zhang Y., Zeng, Z., Sheng, M., **Dang, R.**, Gu, Y., Ding, J., Zhu, L., and Sun, K.: Resolving diurnal point-source ammonia emissions from fertilizer plants using complementary polar-orbiting and geostationary satellite observations, submitted to *Environmental Science & Technology*.

First-authored publications:

2. **Dang, R.**, Jacob, D. J., He, M., Clifton, O. E., Li, K., and Liao, H.: Lightning source of nitrogen oxides inferred from geostationary satellites: global implications for oxidant chemistry, https://doi.org/10.1029/2026GL124557, *Geophysical Research Letters,* 2026.
3. **Dang, R.**, Jacob, D. J., Wang, H., Nowlan, C. R., Gonzalez Abad, G., Chong, H., Liu, X., Shah, V., Yang, L. H., Oak, Y. J., Marais, E. A., Horner, R. P., Rollins, A. W., Crawford, J. H., Li, K., and Liao, H.: High-

resolution geostationary satellite observations of free tropospheric NO_2 over North America: implications for lightning emissions, https://doi.org/10.1073/pnas.2510535122, *Proceedings of the National Academy of Sciences*, 2025.

4. **Dang, R**., Jacob, D. J., Zhai, S., Yang, L. H., Pendergrass D. C., Coheur, P., Clarisse, L., Van Damme, M., Choi, J., Park, J., Liu, Z., Xie, P., and Liao, H.: A satellite-based indicator for diagnosing particulate nitrate sensitivity to precursor emissions: application to East Asia, Europe, and North America, https://doi.org/10.1021/acs.est.4c08082, *Environmental Science & Technology*, 2024.
5. **Dang, R.**, Jacob, D. J., Zhai, S., Coheur, P., Clarisse, L., Van Damme, M., Pendergrass, D. C., Choi, J.-s., Park, J.-s., Liu, Z., and Liao, H.: Diagnosing the Sensitivity of Particulate Nitrate to Precursor Emissions Using Satellite Observations of Ammonia and Nitrogen Dioxide, https://doi.org/10.1029/2023GL105761, *Geophysical Research Letters*, 2023.
6. **Dang, R.**, Jacob, D. J., Shah, V., Eastham, S. D., Fritz, T. M., Mickley, L. J., Liu, T., Wang, Y., and Wang, J.: Background nitrogen dioxide (NO_2) over the United States and its implications for satellite observations and trends: effects of nitrate photolysis, aircraft, and open fires, https://doi.org/10.5194/acp-23-6271-2023, *Atmospheric Chemistry and Physics*, 2023.
7. **Dang, R.**, Liao, H., and Fu, Y.: Quantifying the anthropogenic and meteorological influences on summertime surface ozone in China over 2012-2017, https://doi.org/10.1016/j.scitotenv.2020.142394, *Science of the Total Environment*, 2021.
8. **Dang, R.**, and Liao, H.: Radiative forcing and health impact of aerosols and ozone in China as the consequence of clean air actions over 2012-2017, https://doi.org/10.1029/2019GL084605, *Geophysical Research Letters*, 2019.
9. **Dang R.**, and Liao, H.: Severe winter haze days in the Beijing-Tianjin-Hebei region from 1985-2017 and the roles of anthropogenic emissions and meteorology, https://doi.org/10.5194/acp-19-10801-2019, *Atmospheric Chemistry and Physics*, 2019.

Co-authored publications:

10. Zheng, Y., Zeng, Z., Zhou, R., Clarisse, L., Clerbaux, C., **Dang, R.**, Wei, J., Tang, M., Liu, M., and Hu, M: Space-based observation of decadal variations of ammonium sulfate aerosols over northern China, https://doi.org/10.1029/2026GL124853, *Geophysical Research Letters,* 2026.
11. Sheng, M., Zhou, R., Hua, J., Han, S., Liu, S., Zhang, L., Wang, W., **Dang, R.**, Cao, H., Chen, Z., et al. Geostationary observations of atmospheric ammonia over East Asia: spatio-temporal variations revealed by three years of FY-4B/GIIRS measurements, https://doi.org/10.5194/acp-26-7803-2026, *Atmospheric Chemistry and Physics*, 2026
12. Yang, L. H., Jacob, D. J., Bates, K. H., Lin, H., Allen, H. M., Muller, J.-F., Brown, S. S., **Dang, R.**, Colombi, N. K., Zhai, S., Yantosca, R. M., Brewer, J. F., Ng, N. L., Crounse, J. D., Wennberg, P. O., Li, K., Liao, H.: Modeling of methyl hydroperoxide observations in urban and remote air over South Korea: methylperoxy radical chemistry and inference of atmospheric methanediol, https://doi.org/10.1029/2025GL118267, *Geophysical Research Letters*, 2025.
13. Pendergrass, D. C., Jacob, D. J., Oak, Y. J., **Dang, R.**, Yang, L. H., Beaudry, E., Zhai, S., Kim, H., Chi, J., Park, J., Kim, S., Li, K., and Liao, H.: Wintertime trends of fine particulate matter ($PM_{2.5}$) in South Korea, 2012-2022: response of nitrate and organic components to decreasing NO_x emissions, https://doi.org/10.1029/2025GL116091, *Geophysical Research Letters*, 2025.
14. Li, B., Liao, H., Li, K., Lin, J., Gong, C., Liu, H., Li, Y., Chen, L., Yang, Y., Jin, X., Zhao, Y., Wang, T., Jin, J., **Dang, R.**, and Jacob, D. J.: Environmental burden and health inequity in China's road-based express delivery, https://doi.org/10.1038/s44284-025-00300-3, *Nature Cities*, 2025.

15. Oak, Y. J., Jacob, D. J., Pendergrass, D. C., **Dang, R.**, Chong, H., Lee, S., Kuk, S., and Kim, J.: Air quality trends and regimes in South Korea inferred from 2015-2023 surface and satellite observations, https://doi.org/10.5194/acp-25-3233-2025, *Atmospheric Chemistry and Physics*, 2025.
16. Yang, L. H., Jacob, D. J., Lin, H., **Dang, R.**, Bates, K. H., East, J. D., Travis, K. R., Pendergrass, D. C., and Murray, L. T.: Assessment of hydrogen's climate impact is affected by model OH biases, https://doi.org/10.1029/2024GL112445, *Geophysical Research Letters*, 2025.
17. Li, B., Liao, H., Li, K., Wang, Y., Zhang, L., Guo, Y., Liu, L., Jin, J., Yang, Y., Gong, C., Wang, T., Shen, W., Wang, P., **Dang, R.**, Liao, K., Zhu, Q., and Jacob, D. J.: Unlocking nitrogen management potential via large-scale farming for air quality and substantial co-benefits, https://doi.org/10.1093/nsr/nwae324, *National Science Review*, 2024.
18. Lin, H., Emmons, L. K., Lundgren, E. W., Yang, L. H., Feng, X., **Dang, R.**, Zhai, S., Tang, Y., Kelp, M. M., Colombi, N. K., Eastham, S. D., Fritz, T. M., and Jacob, D. J.: Intercomparison of GEOS-Chem and CAM-chem tropospheric oxidant chemistry within the Community Earth System Model version 2 (CESM2), https://doi.org/10.5194/acp-24-8607-2024, *Atmospheric Chemistry and Physics*, 2024.
19. Yang, L. H., Jacob, D. J., **Dang, R.**, Oak, Y. J., Lin, H., Kim, J., Zhai, S., Colombi, N. K., Pendergrass, D. C., Beaudry, E., Shah, V., Feng, X., Yantosca, R. M., Chong, H., Park, J., Lee, H., Lee, W. J., Kim, S., Kim, E., Travis, K. R., Crawford, J. H., and Liao, H.: Interpreting GEMS geostationary satellite observations of the diurnal variation of nitrogen dioxide (NO_2) over East Asia, https://doi.org/10.5194/acp-24-7027-2024, *Atmospheric Chemistry and Physics*, 2024.
20. Zhai, S., Jacob, D. J., Franco, B., Clarisse, L., Coheur, P., Shah, V., Bates, K. H., Lin, H., **Dang, R.**, Sulprizio, M. P., Huey, L. G., Moore, F. L., Jaffe, D. A., and Liao, H.: Transpacific transport of Asian peroxyacetyl nitrate (PAN) observed from satellite: implications for ozone, https://doi.org/10.1021/acs.est.4c01980, *Environmental Science & Technology*, 2024.
21. Shah, V., Jacob, D. J., **Dang, R.**, Lamsal, L. N., Strode, S. A., Steenrod, S. D., Boersma, K. F., Eastham, S. D., Fritz, T. M., Thompson, C., Peischl, J., Bourgeois, I., Pollack, I. B., Nault, B. A., Cohen, R. C., Campuzano-Jost, P., Jimenez, J. L., Andersen, S. T., Carpenter, L. J., Sherwen, T., and Evans, M. J.: Nitrogen oxides in the free troposphere: implications for tropospheric oxidants and the interpretation of satellite NO_2 measurements, https://doi.org/10.5194/acp-23-1227-2023, *Atmospheric Chemistry and Physics*, 2023.
22. Li, J., Hao, X., Liao, H., Dai, H., Li, N., Gu, Y., **Dang, R.**, Li, B., and Wei, Y.: Air pollution mitigation in North China through flexible heating policies, https://doi.org/10.1088/1748-9326/acb3e2, *Environmental Research Letter*, 2023.
23. Wang, P., Yang, Y., Li, H., Chen, L., **Dang, R.**, Xue, D., Li, B., Tang, J., Leung, L. R., and Liao, H.: North China Plain as a hot spot of ozone pollution exacerbated by extreme high temperatures, https://doi.org/10.5194/acp-22-4705-2022, *Atmospheric Chemistry and Physics*, 2022.
24. Hao, X., Li, J., Wang, H., Liao, H., Yin, Z., Hu, J., Wei, Y., and **Dang, R.**: Long-term health impact of $PM_{2.5}$ under whole-year COVID-19 lockdown in China, https://doi.org/10.1016/j.envpol.2021.118118, *Environmental Pollution*, *2021.*
25. Gong, C., Liao, H., Zhang, L., Yue, X., **Dang, R.**, and Yang, Y.: Persistent ozone pollution episodes in North China exacerbated by regional transport, https://doi.org/10.1016/j.envpol.2020.115056, *Environmental Pollution*, 2020.
26. Zhu, J., Chen, L., Liao, H., and **Dang, R.**: Correlations between $PM_{2.5}$ and Ozone over China and Associated Underlying Reasons, https://doi.org/10.3390/atmos10070352, *Atmosphere*, 2019.

PRESENTATIONS

2026	Global Change Remote Sensing Seminar, PKU	Beijing, China (Invited talk)
2026	The 12th International GEOS-Chem Meeting	St. Louis, MO (Talk)
2025	AGU Fall Meeting	New Orleans, LA (Poster)
2025	The TEMPO/GeoXO ACX Joint Science Team Workshop	Boston, MA (Talk)
2025	NOAA CSL AGES+ Seminar	Virtual (Invited talk)
2025	NASA GMAO GEOS-CF Meeting	Virtual (Talk)
2025	TEMPO Science Team Telecon	Virtual (Talk)
2024	AGU Fall Meeting	San Francisco, CA (Talk)
2024	Nanjing University, School of Atmospheric Sciences Seminar	Nanjing, China (Invited talk)
2024	The 30th Atmospheric Environment Meeting	Hefei, China (Talk)
2024	Nanjing University of Information Science & Technology, School of Environmental Science and Engineering Seminar	Nanjing, China (Invited talk)
2024	Nankai University, College of Environmental Science and Engineering Seminar	Tianjin, China (Invited talk)
2024	The TEMPO/GEMS Joint Science Team Workshop	Kailua-Kona, HA (Talk)
2024	The 11th International GEOS-Chem Meeting	St. Louis, MO (Talk)
2023	AGU Fall Meeting	San Francisco, CA (Poster)
2023	Joint Science Meeting for TEMPO, GeoXO ACX, & TOLNet	Huntsville, AL (Poster)
2023	AMS Annual Meeting	Denver, CO (Talk)
2022	AGU Fall Meeting	Chicago, IL (Poster)
2022	The 10th International GEOS-Chem Meeting	St. Louis, MO (Talk)
2021	Peking University, Department of Atmospheric and Oceanic Sciences Seminar	Virtual (Invited talk)
2020	The 26th Atmospheric Environment Meeting	Virtual (Talk)
2019	AGU Fall Meeting	San Francisco, CA (Talk)
2019	The 25th Atmospheric Environment Meeting	Chengdu, China (Talk)
2019	The 9th International GEOS-Chem Meeting	Cambridge, MA (Poster)
2018	AGU Fall Meeting	Washington D. C. (Talk)
2018	The 24th Atmospheric Environment Meeting	Qingdao, China (Talk)
2018	The 1st Regional GEOS-Chem Asia Meeting	Nanjing, China (Talk)
2017	The 8th International GEOS-Chem Meeting	Cambridge, MA (Poster)

TEACHING & MENTORING EXPERIENCE

Teaching Assistant 2018 – 2019

Aerosols and Climate Change, University of Chinese Academy of Sciences

Co-advisor July 2025 – Present

Wei Zhou (visiting graduate student, NUIST)
Project: Impact of agricultural emissions on climate system
Shuxin Li (visiting master student, NUIST)
Project: Interactions between anthropogenic emissions and climate change
Liubin Pan (visiting master student, NUIST)
Project: Drivers of extreme ozone pollution in China

Mentor 2023 – 2025
Yunxiao Tang (graduate student, Harvard University)
Project: CHEEREIO inversion of TEMPO NO_2 to infer NO_x emissions

Mentor Summer 2024
Jaden Southern (visiting undergraduate student, Stanford University)
Project: Evaluating TEMPO HCHO and NO_2 products

SELECTED AWARDS & HONORS

Outstanding Student Presentation Award, *Chinese Society for Environmental Sciences* 2018

Outstanding Student Award, *Chinese Academy of Sciences* 2016

Outstanding Graduate Award, *Nanjing University* 2015

Outstanding Student Leadership Award, *Nanjing University* 2014 – 2015

National Scholarship, *Nanjing University* 2014

Outstanding Student Award, *Nanjing University* 2012 – 2015

People's Scholarship, *Nanjing University* 2012 – 2013

SERVICE

Proposal reviewer NASA (2025)

Peer Reviewer *One Earth, Environmental Science & Technology Air, Environmental Science & Technology Letters, Atmospheric Chemistry & Physics, Geophysical Research Letters, Remote Sensing of Environment, Earth System Science Data, Geoscientific Model Development, Environmental Research Letters, Journal of Geophysical Research – Atmospheres, Earth's Future, Environment International*

Co-Chair TEMPO/GeoXO ACX Joint Science Team Workshop, Science and Applications Session (2025)

Leader *Air Quality & Chemistry subgroup*, Harvard Atmospheric Chemistry Modeling Group (2022 – present)

Organizer Harvard Atmospheric & Environmental Chemistry Seminars (2022 – 2023)

Participant TEMPO Science Team Meeting on Validation of Satellite Products (November 2023 – April 2025)

Member American Geophysical Union, American Meteorological Society

Volunteer OSPA Judge at AGU Fall Meeting (2023, 2024)